Apertus 1.5 under test

Academic benchmarks and real business tasks



Prof. Dr. Siegfried Handschuh
Götz-Henrik Wiegand
DS-NLP Lab · University of St.Gallen
«Hack Apertus» Online Session · 2 October 2026

University of St.Gallen
Institute for Computer Science
NLP LAB
Chair for Data Science and Natural Language Processing

Overiew

01 

ACADEMIC BENCHMARKS

Apertus 1.5 against 1.0:
text, 14 languages, reasoning, thinking mode.

02 

BUSINESS TASKS

Six real business benchmarks, 10 models, plus a vision and an audio test: where Apertus shines and where it slips.

03 

YOUR HACK

What does it mean for you?

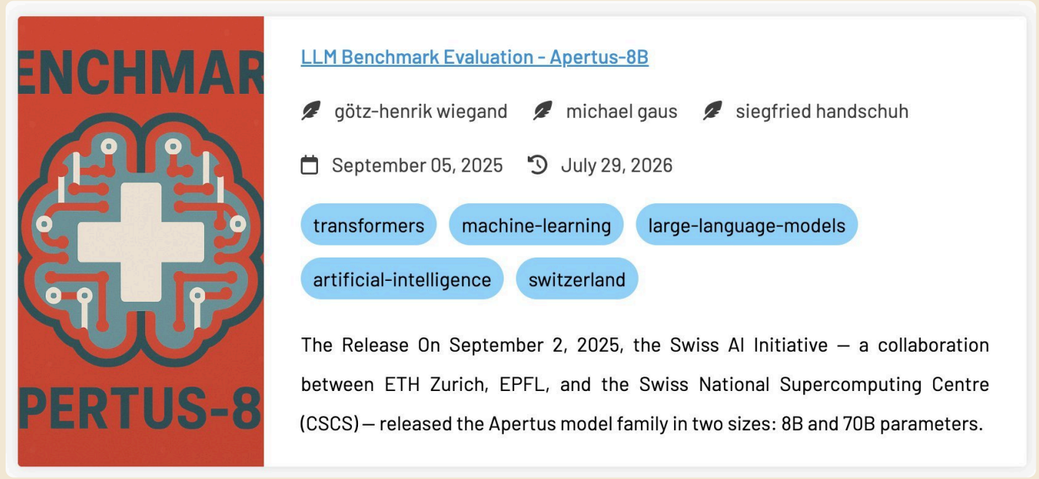
How to avoid Pitfalls?

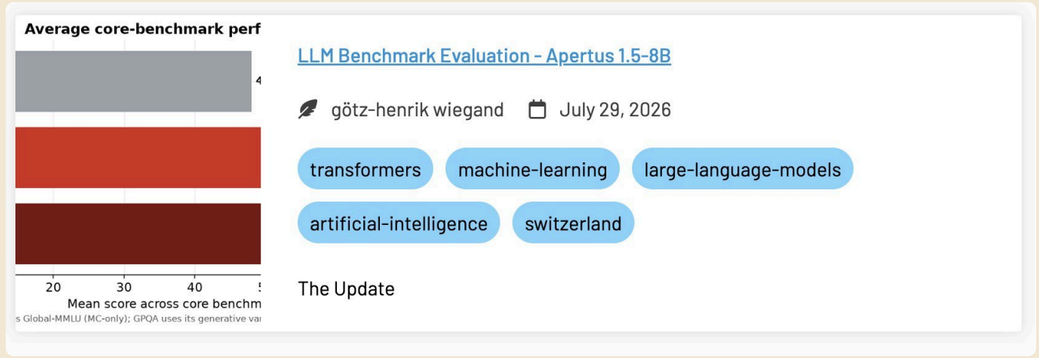
Part 1 · Academic benchmarks

Götz-Henrik Wiegand

Our Apertus benchmarks, then and now

Independent evaluation at the DS-NLP chair, University of St.Gallen





2025 · Apertus 1.0

First independent benchmark study

Comparison with other small models

Blog post 2025

2026 · Apertus 1.5 8B

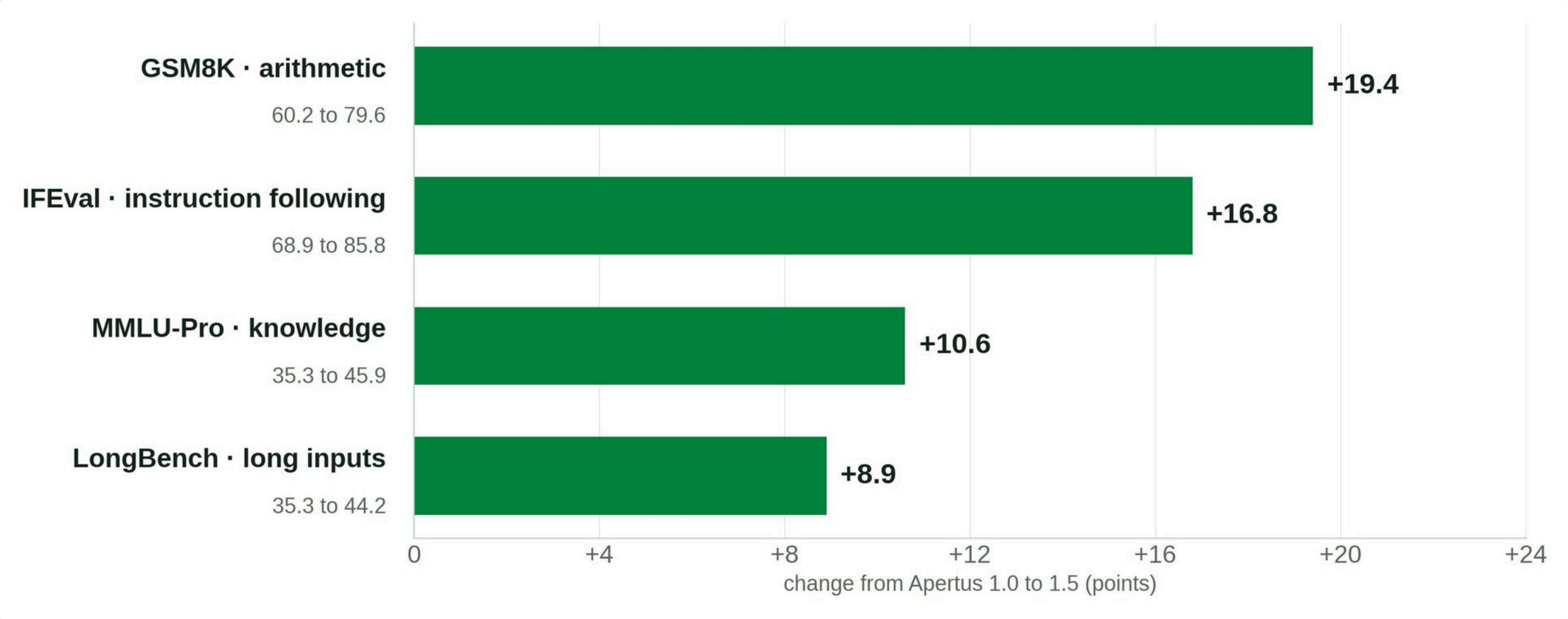
1.0 baseline re-measured in the current setup

Normal mode, thinking and image input

Blog post 2026

1.5 improves across the core text tests

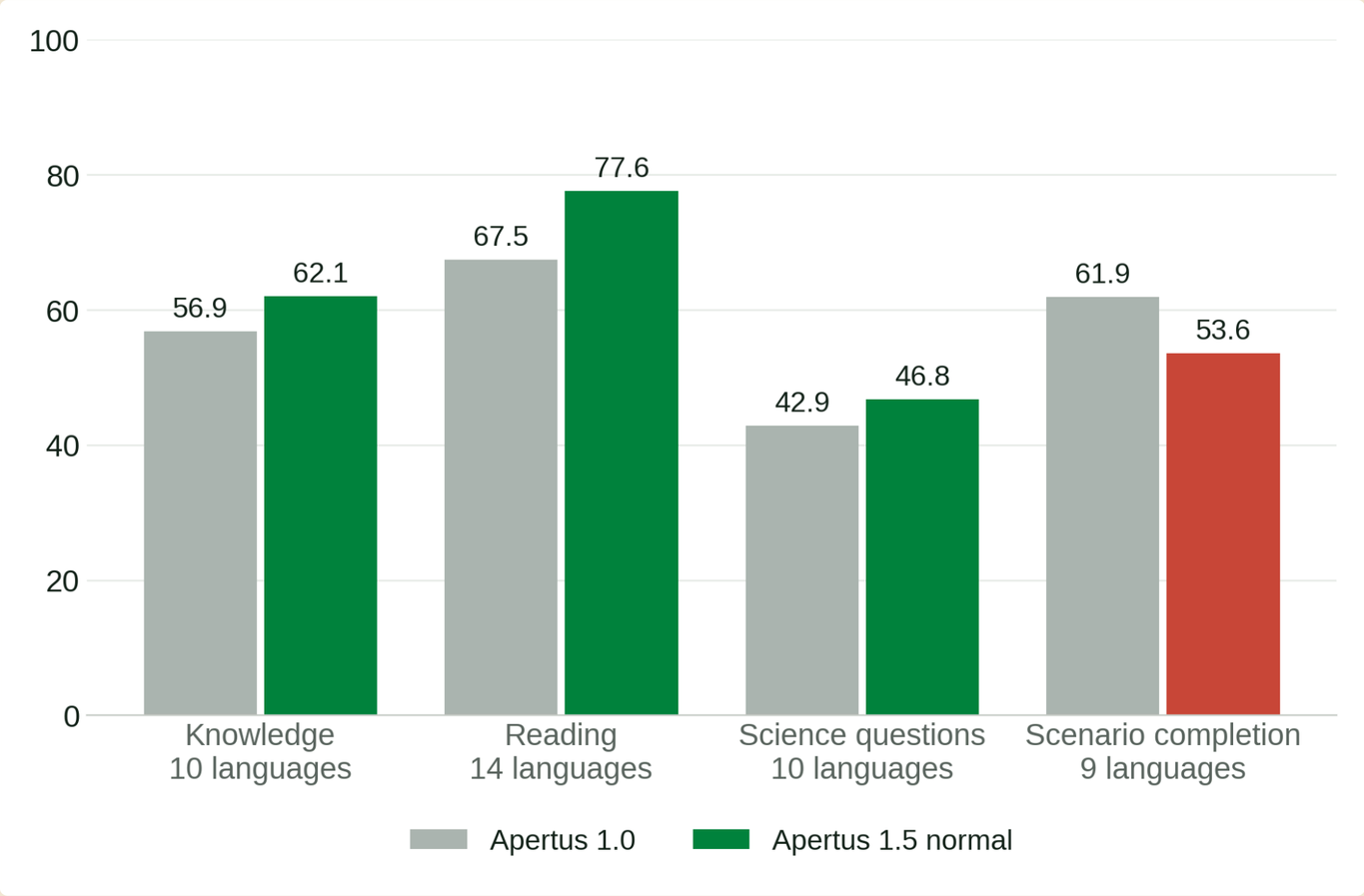
Apertus 1.0 and 1.5 in normal mode, using our current evaluation setup



Gains use unrounded results. Percentage points, except LongBench score points. [1]

Multilingual gains, with one regression

Family averages across the languages we tested, normal mode



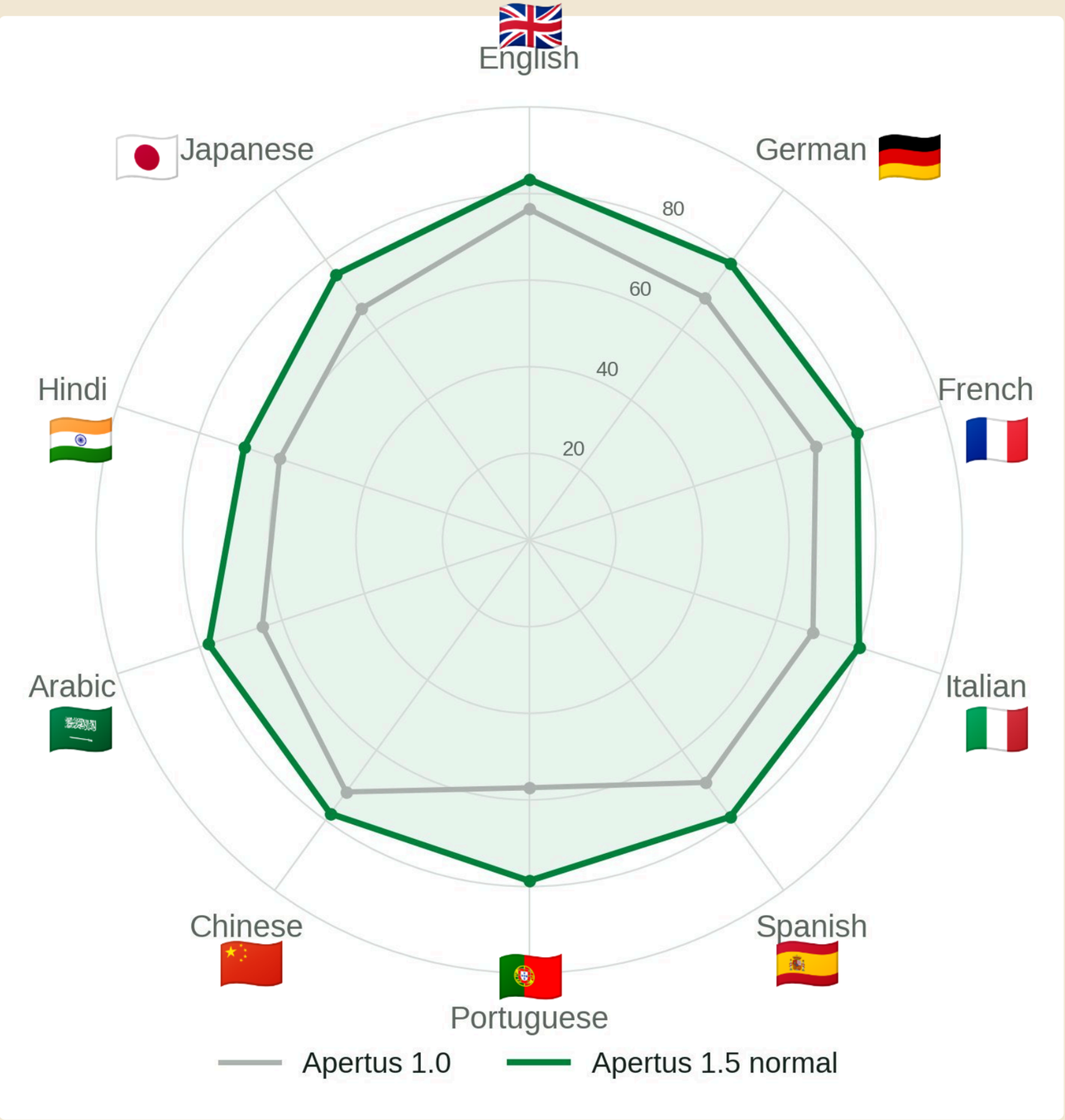
+10.1

points on multilingual reading comprehension

−8.3

points on plausible scenario completion

Accuracy (%). HellaSwag uses length-normalized accuracy. Families use different tasks and language sets. [1]



Reading improves in every language

14 / 14

tested languages improve

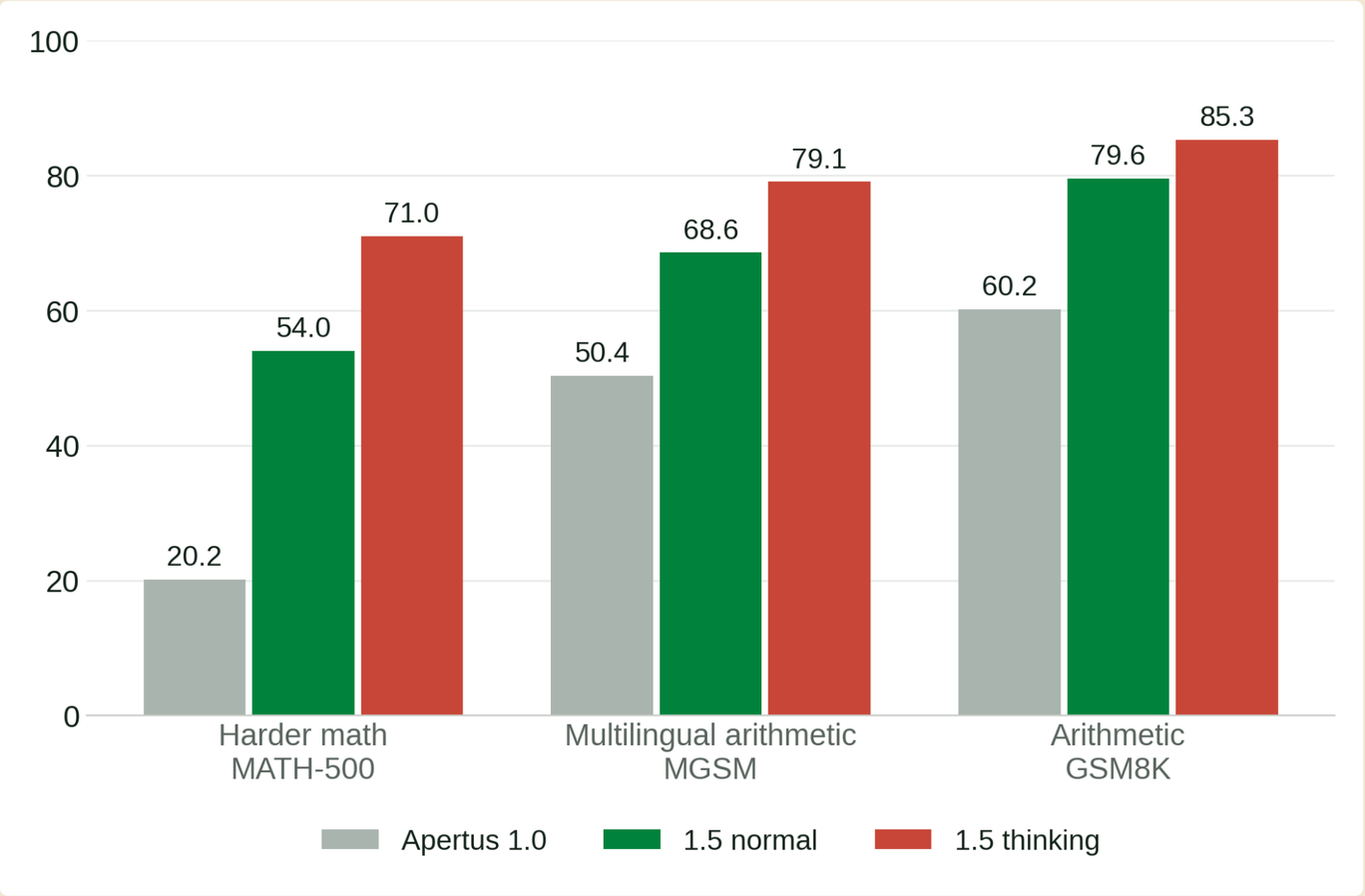
+21.4

points for Portuguese, the largest gain

Belebele reading accuracy (%), scale 0 to 100. Not in the plot: Dutch, Polish, Russian and Turkish. [1]

Thinking helps most in mathematics

Correct-answer accuracy (%) on three math benchmarks



+17.0

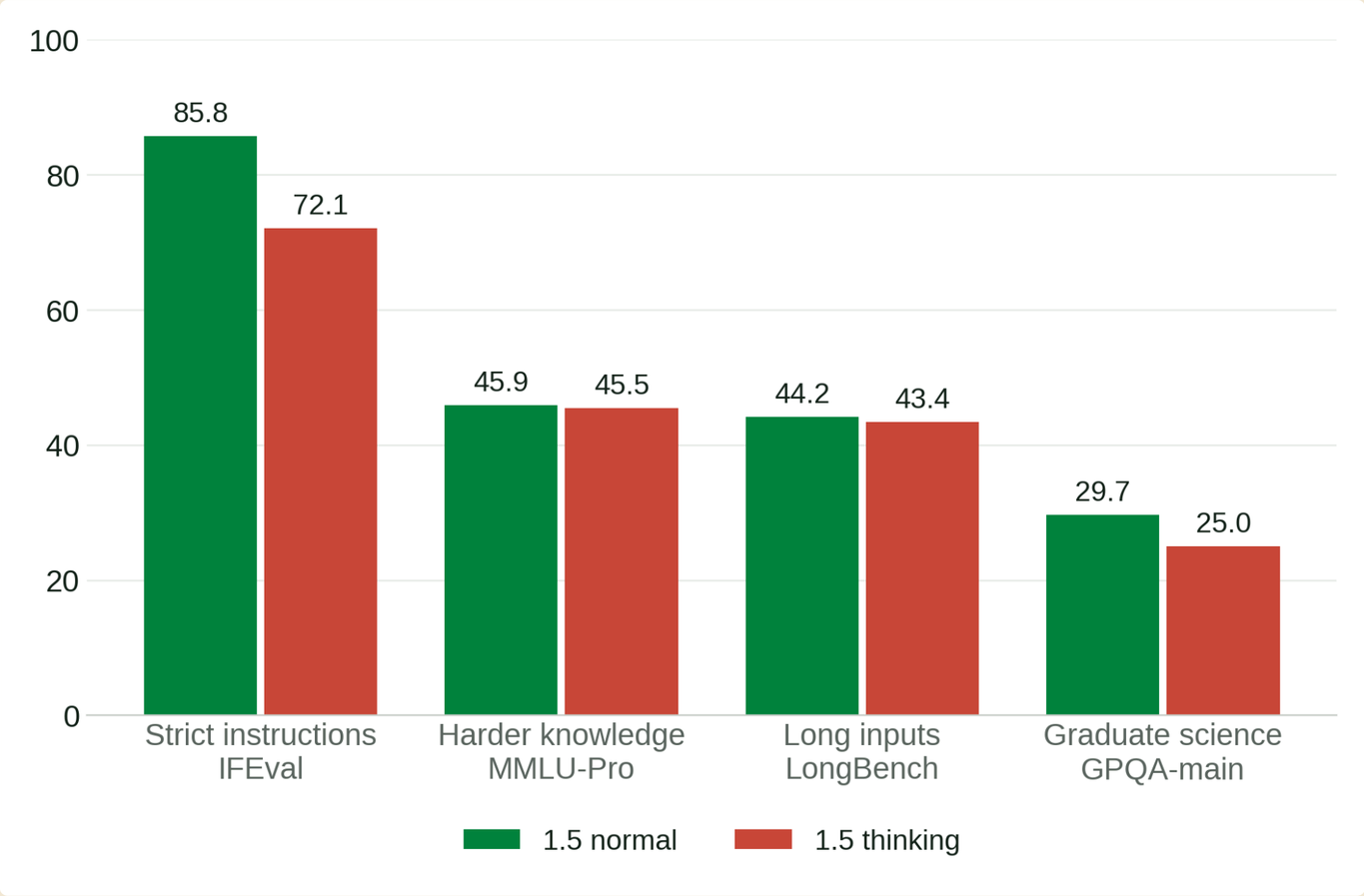
points from thinking on MATH-500

A useful option for math-heavy projects.

MATH-500 uses symbolic answer verification. Reasoning budgets: 4,096 tokens normal, 16,384 thinking. [1]

Thinking can hurt instruction following

Apertus 1.5 normal mode compared with thinking mode



−13.7

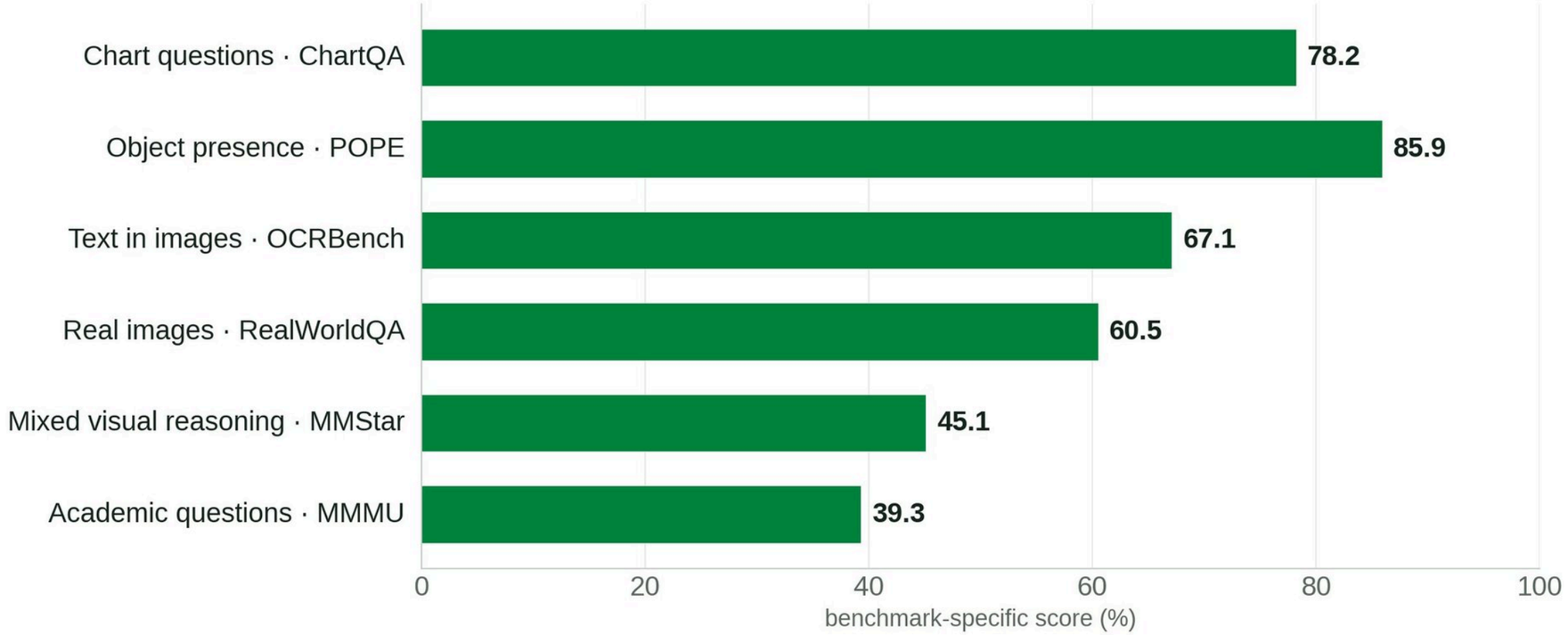
points on strict instruction following

Normal mode is a useful starting point for constrained output.

Accuracy (%), except LongBench score. GPQA-main uses the same generative variant for both modes. [1]

Vision reads charts and text well

Apertus 1.5 8B, six full vision test sets



Promising uses

Chart reading

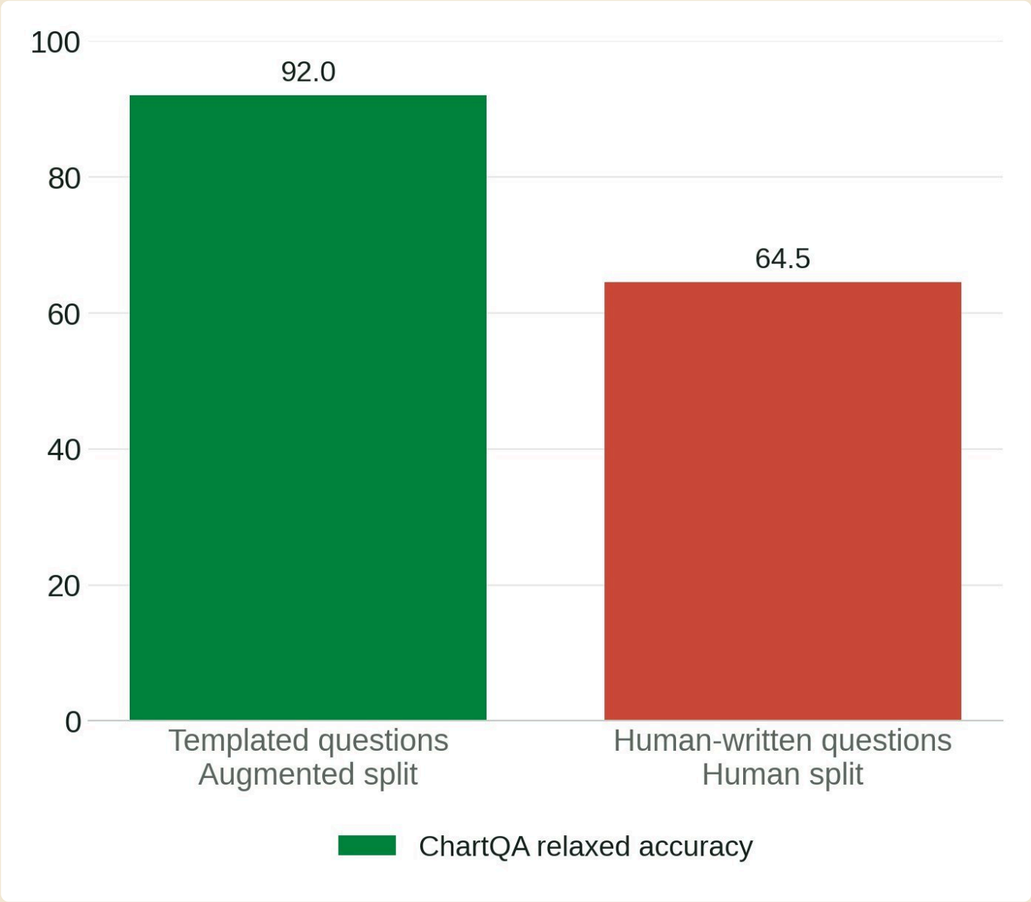
Text extraction

Questions about images

Benchmark-specific scores (%). Different tasks and metrics. No 1.0 vision baseline. [1]

Vision still needs verification

ChartQA reveals a large gap between question types



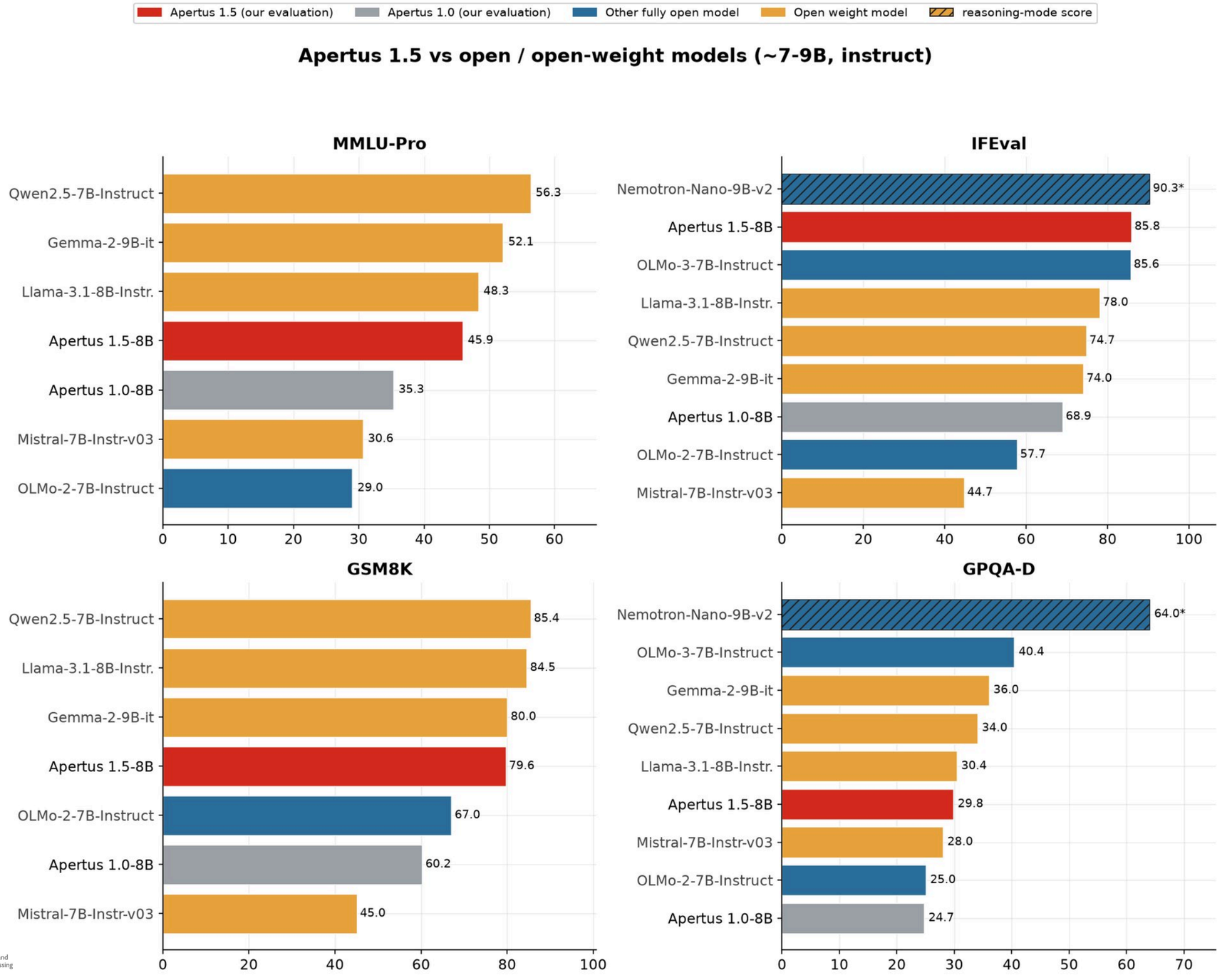
Hard visual reasoning

24.7 % on MMStar science and technology.

Object presence

POPE tests a narrow skill. 79.0 % recall means some objects go undetected.

ChartQA relaxed accuracy (%). Visible evidence and number checks help users verify image answers. [1]



Apertus = our measured results. OLMo-3 & Nemotron from their model cards (Nemotron Nano 2 numbers are reasoning-on, hatched); other models are approximate published numbers (*). Different harnesses and modes, so this is indicative positioning, not a like-for-like leaderboard. Fully open (data+weights+recipe): Apertus, OLMo-3, OLMo-2, Nemotron Nano 2.

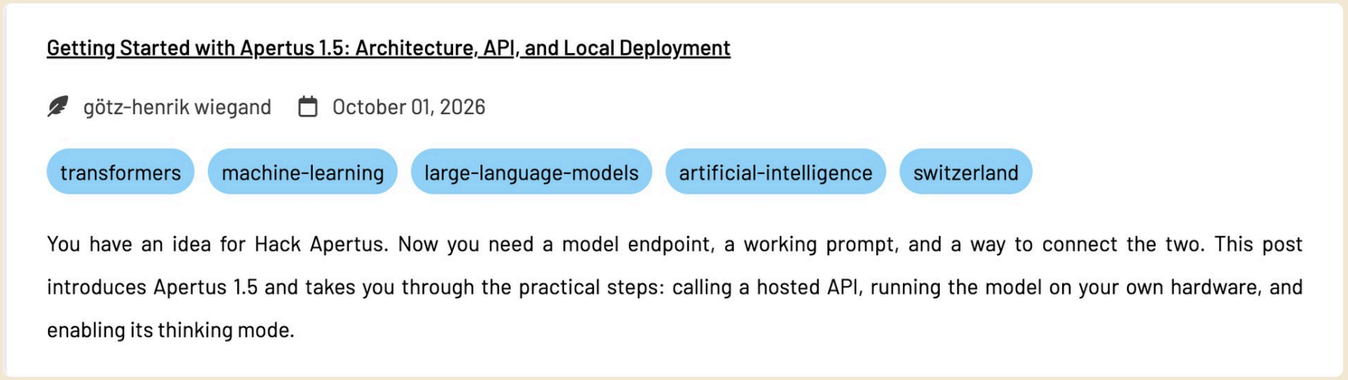
How to get started with Apertus

Practical setup for your own benchmarks and HackApertus projects

Blog post: Getting Started with Apertus

- Tips and tricks
- Quick start for local hosting
- Relevant links and tools
- And more

blog.nlp-lab.ai: Getting Started with Apertus 1.5



Key takeaways from Part 1

Promising capabilities

- Clear gains over Apertus 1.0
- Reading improves in all 14 languages
- Thinking helps multi-step math
- Useful chart and document reading

Limits to keep in mind

- Commonsense completion regresses
- Thinking adds tokens and format risks
- Hard visual reasoning remains weak
- Untested languages need validation

Test your own use case! Next: how does Apertus 1.5 handle real business tasks?

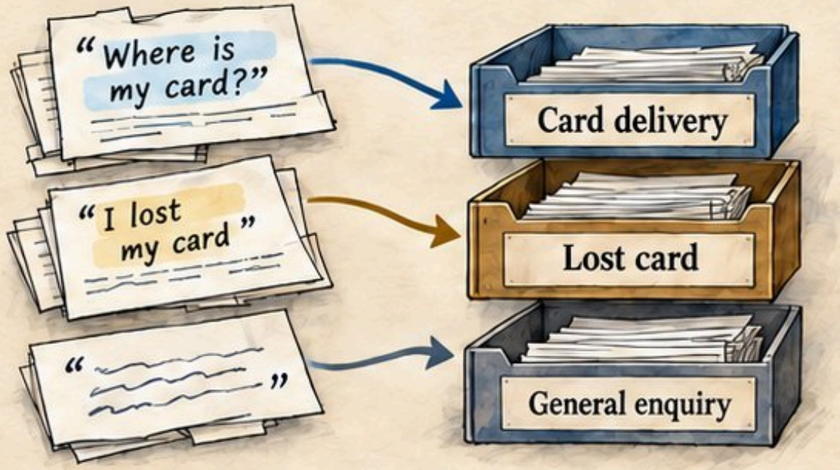
https://blog.nlp-lab.ai/2026/07/29/Apertus15Bench.html

Part 2 · Real business tasks

Six benchmarks, 3,000 real cases per model

Six Office Benchmarks

SWITCHBOARD
Banking77 [2]
What does the customer want?



RECEIPTS
SROIE [3]
What is on this receipt?



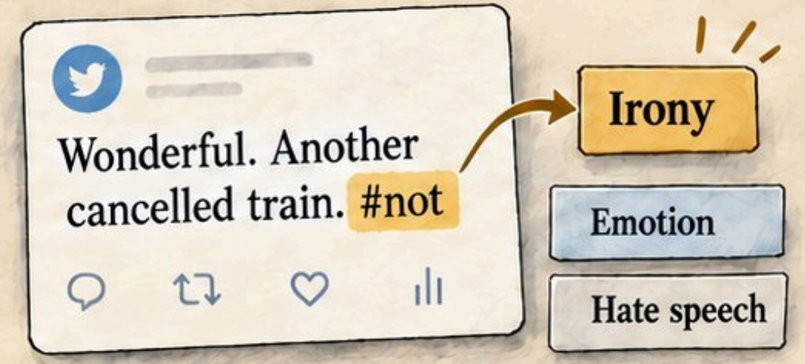
LEGAL
CUAD [4]
What does this clause do?



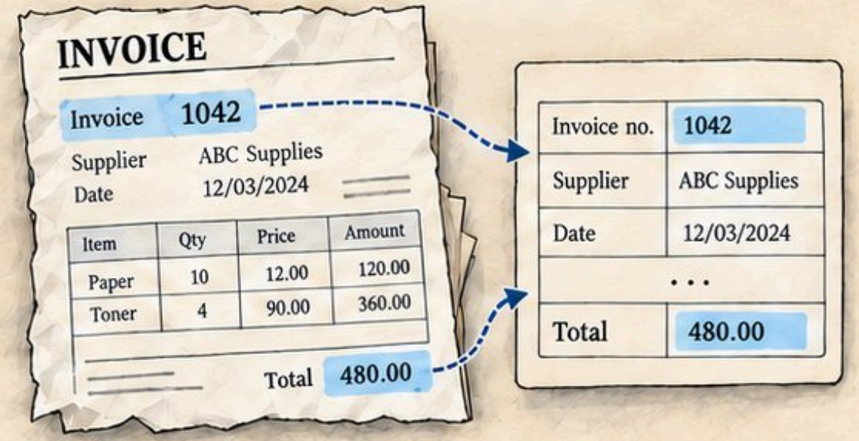
ODD JOBS
RAFT [5]
Eleven real office jobs



SOCIAL MEDIA
TweetEval [6]
Irony and hate in tweets



ACCOUNTS PAYABLE
DocILE [7]
Fill in the invoice form



TESTED

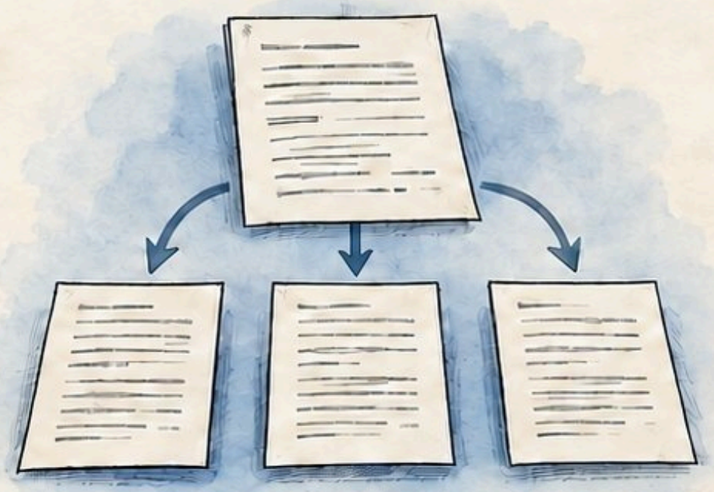


Six public benchmarks in English

About 3,000 cases per model

Plus vision: 361 scanned receipts and audio: 1,200 phone calls

SAME FOR ALL



One zero-shot prompt per task

No prompt tuning

Thinking off





Thinking only in a separate check on hard cases

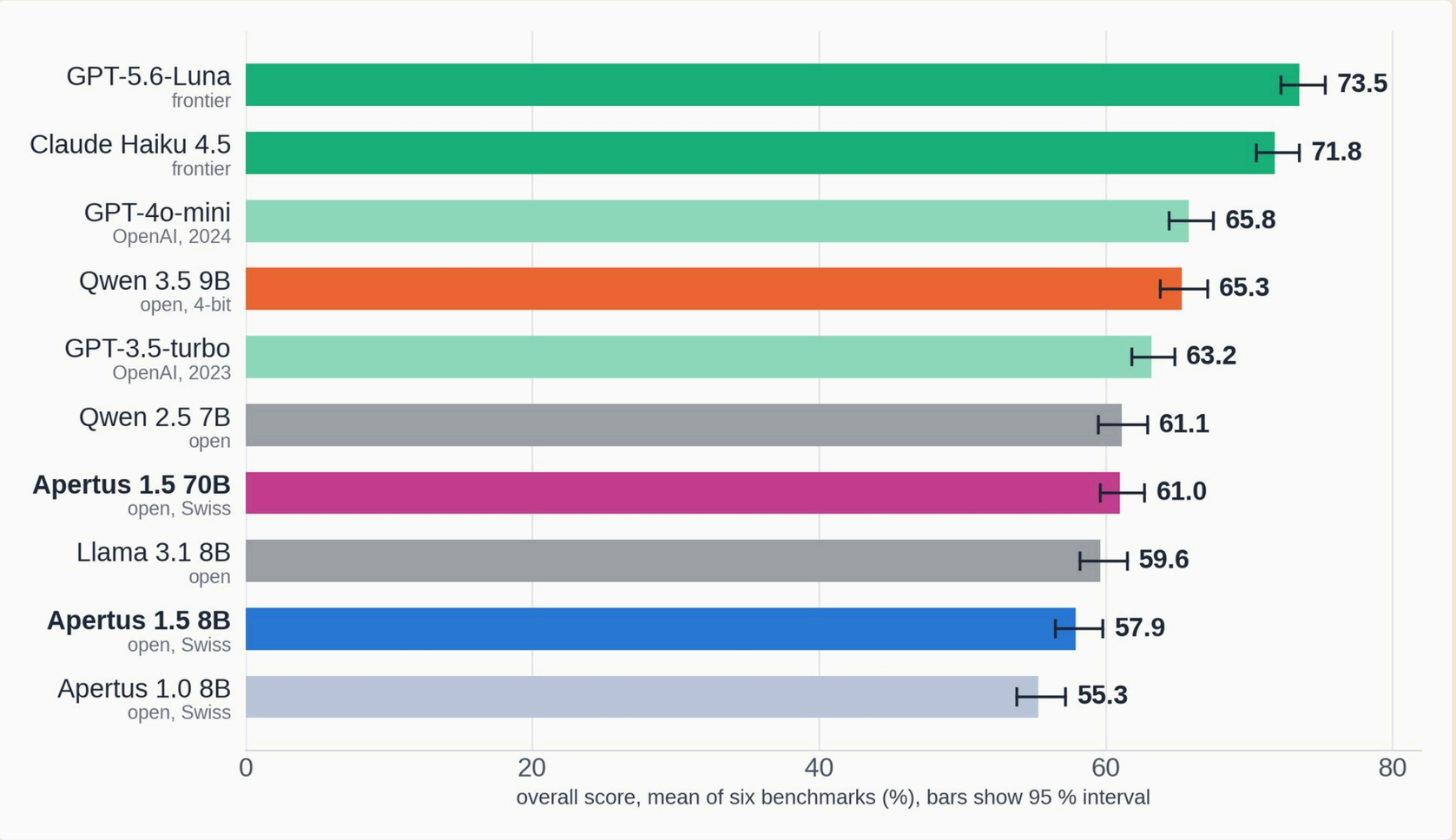
NOT TESTED



Images and audio: one dataset each

Good prompts, a few examples and output checks will move these numbers. That is your room to play.

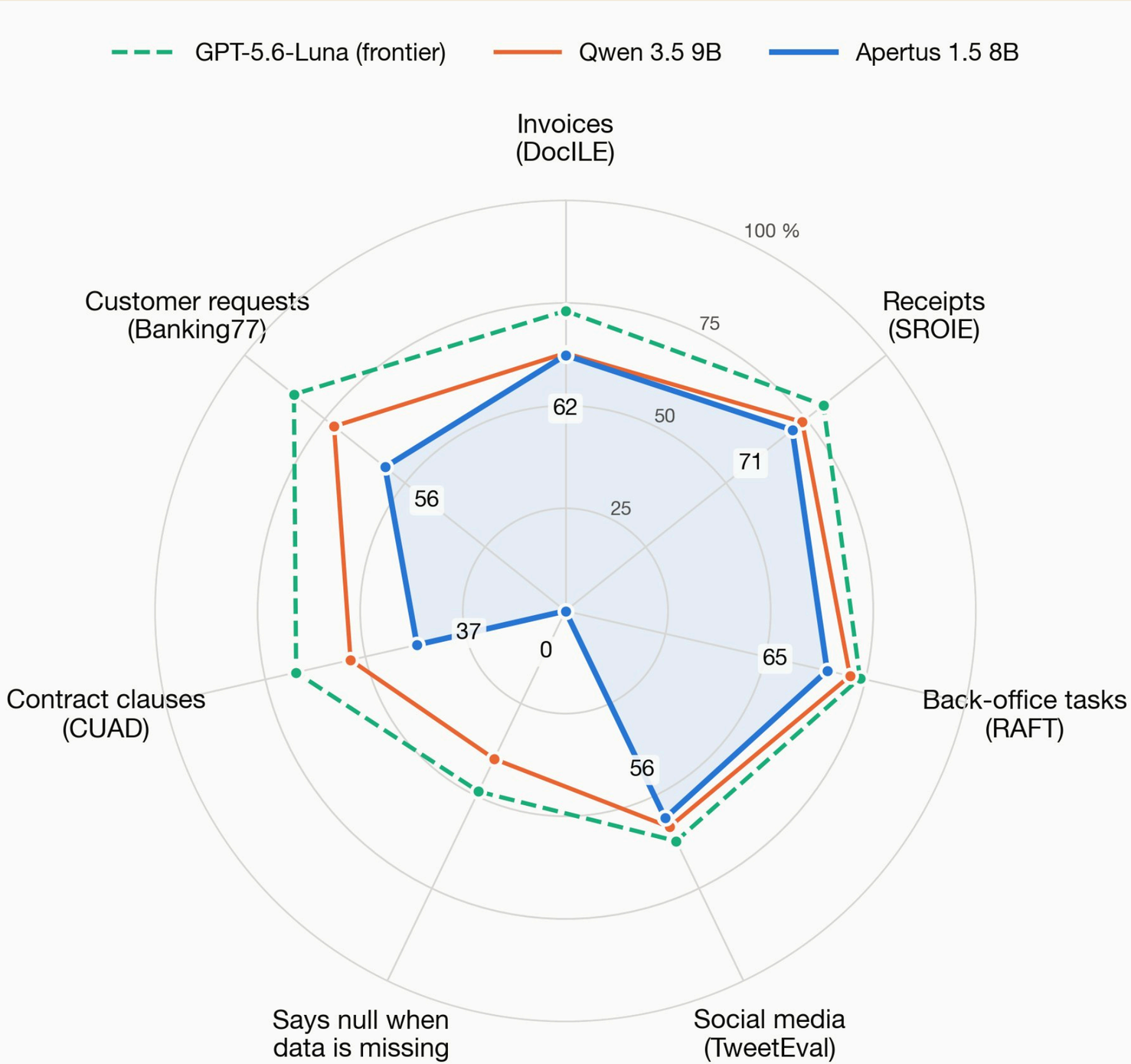
Frontier leads, Apertus 8B holds its own [8]



Apertus 1.5 8B

invoices (DocILE): **62**

contracts (CUAD): **37**



Apertus 1.5 8B at a glance [8]

Easy

DOCUMENTS, SHORT REQUESTS

Dates and amounts on invoices and receipts. Clear customer requests.

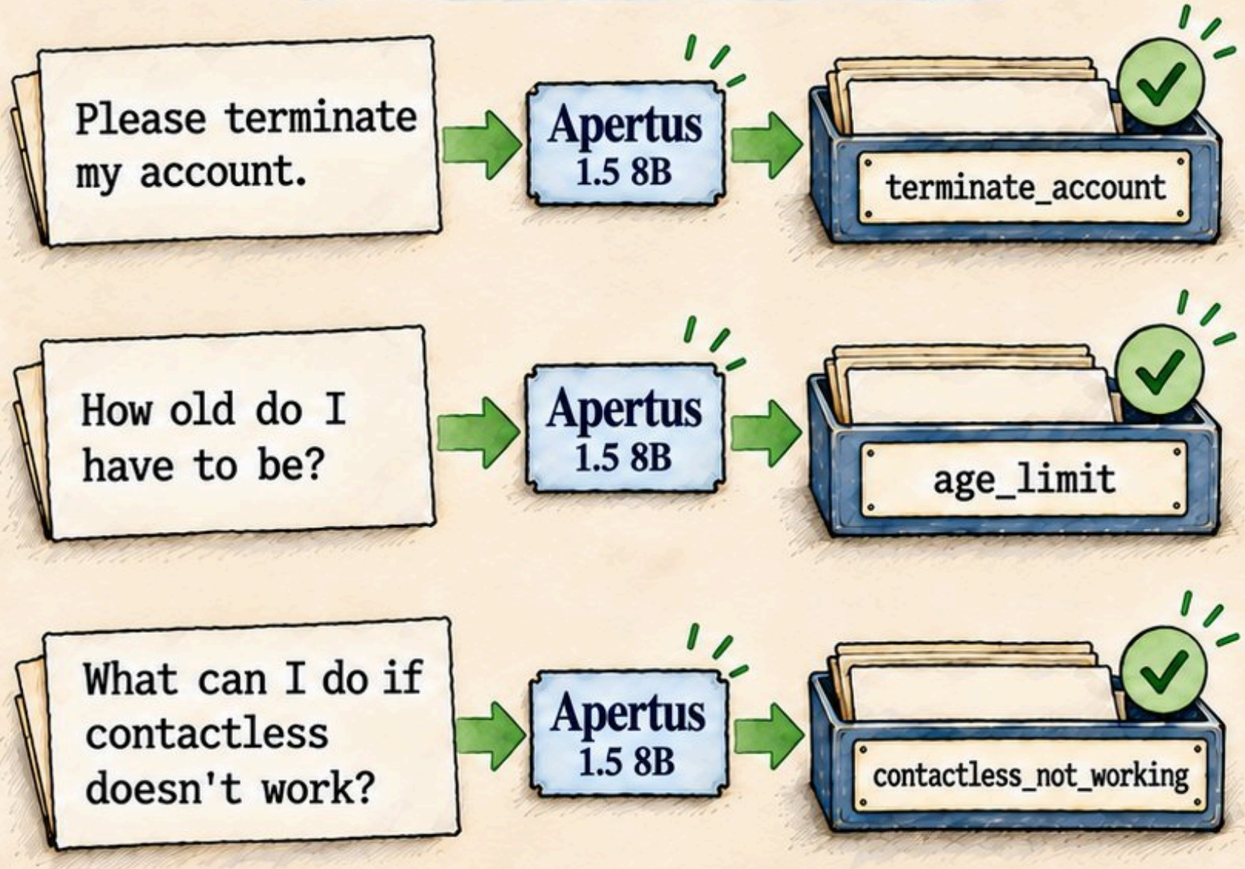
Hard

LONG LISTS, MISSING DATA

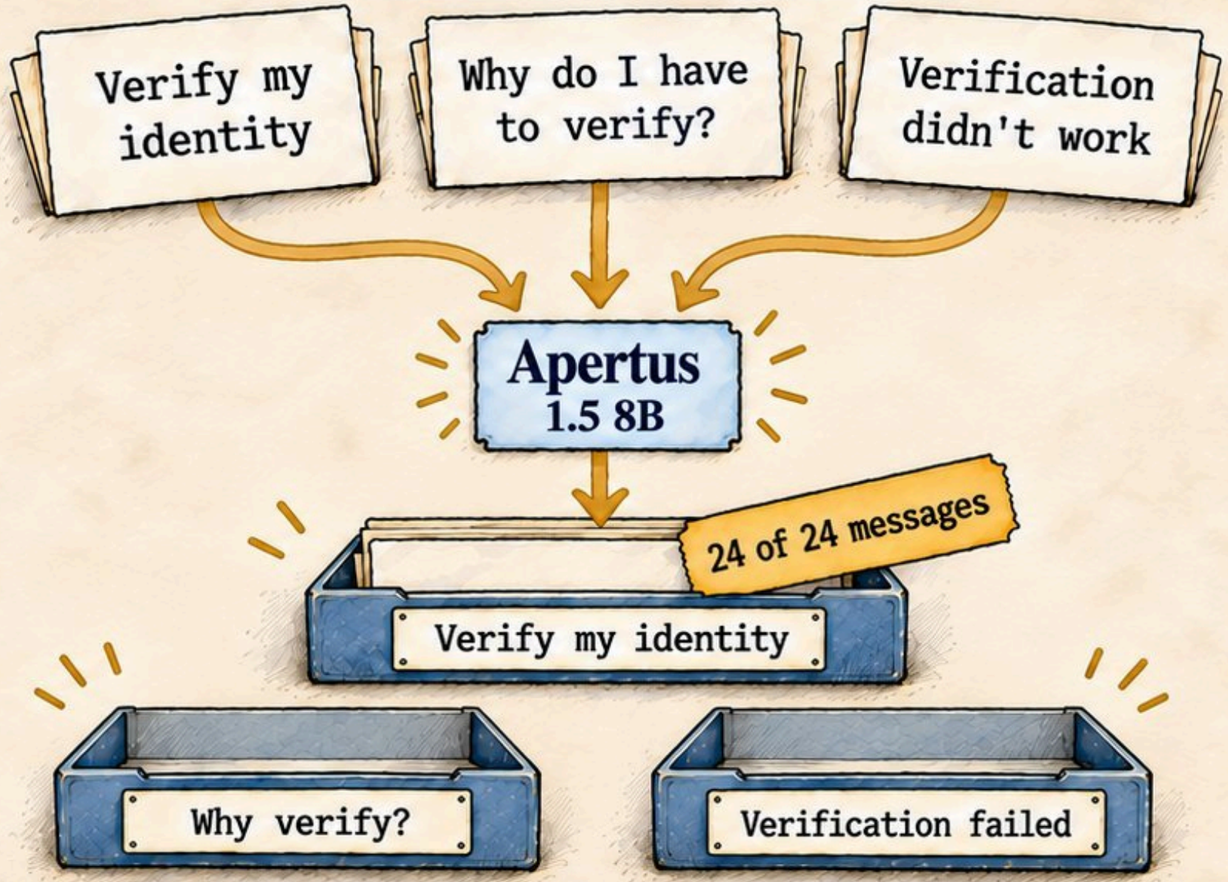
41 contract types, look-alike requests, irony, and saying «null» when something is not there.

Model tested: Apertus 1.5 8B

Routed correctly



Look-alike requests



For your hack: Apertus 1.5 8B routes distinct requests reliably. Merge look-alike categories, or add one example each.



Apertus 1.5 8B read

RIGHT
date: 08/06/2020

RIGHT
total: 3,485.00

WRONG
vendor: the buyer in the letterhead

WRONG
order number: “Purchase Order”

For your hack: tell Apertus which document type it reads, and check who is who with a rule.

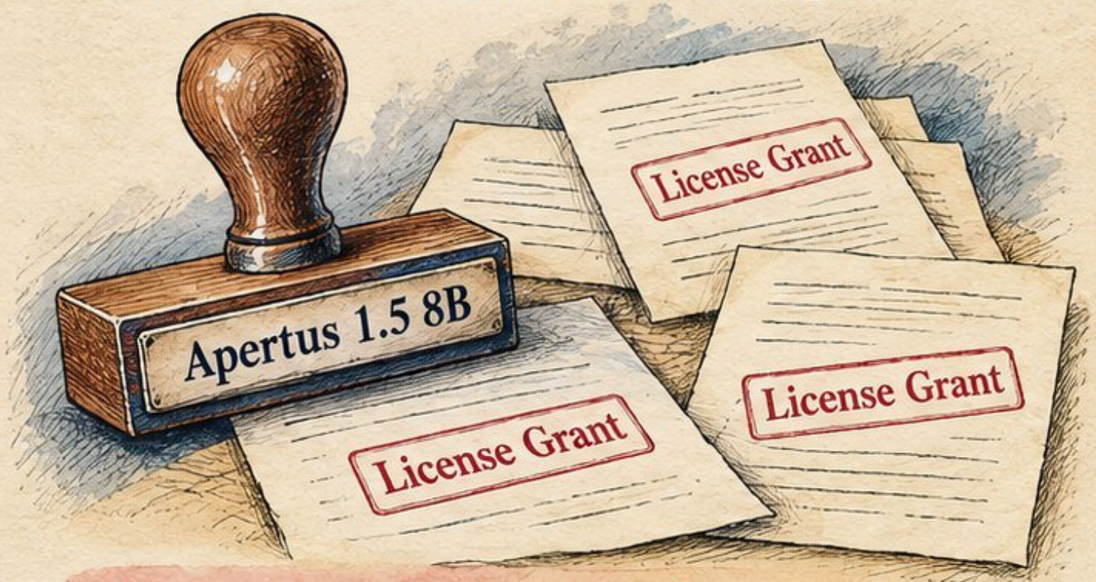
A favourite label

294 contract clauses · 41 types

Contract clause

"The Reseller shall be entitled to enter into agreements with its subsidiaries and affiliates to act as sub-distributors and/or selling agents of the Products in the Territory."

Expected	Affiliate License-Licensee	✓
Apertus 1.5 8B	License Grant	✗
Apertus 1.5 70B	Affiliate License-Licensee	✓



62 answers: "License Grant"

Only 6 actual "License Grant" clauses

Invented labels:
"Severability" · "Limited Liability" ✗

For your hack: with 40+ categories, classify in two steps, or check every answer against the list.

Just paid $2.59 for gas!
#ThanksObama #sarcasm

Full on man flu just in time for a weekend of work!!
#awesome #sarcasm

Allowed labels	irony	non_irony
Apertus 1.5 8B	sarcasm	✗
Apertus 1.5 70B	irony	✓

9 of 150 tweets



The meaning is right,
the word is not on the list.
Every one counts as wrong.

For your hack: map synonyms in code,
or constrain the output to the allowed labels.

Benchmark test case

Source excerpt

GARDENIA BAKERIES (KL) SDN BHD
TAX INVOICE / ADJUSTMENT NOTE
CASH INV NO.: 7901F709 ...

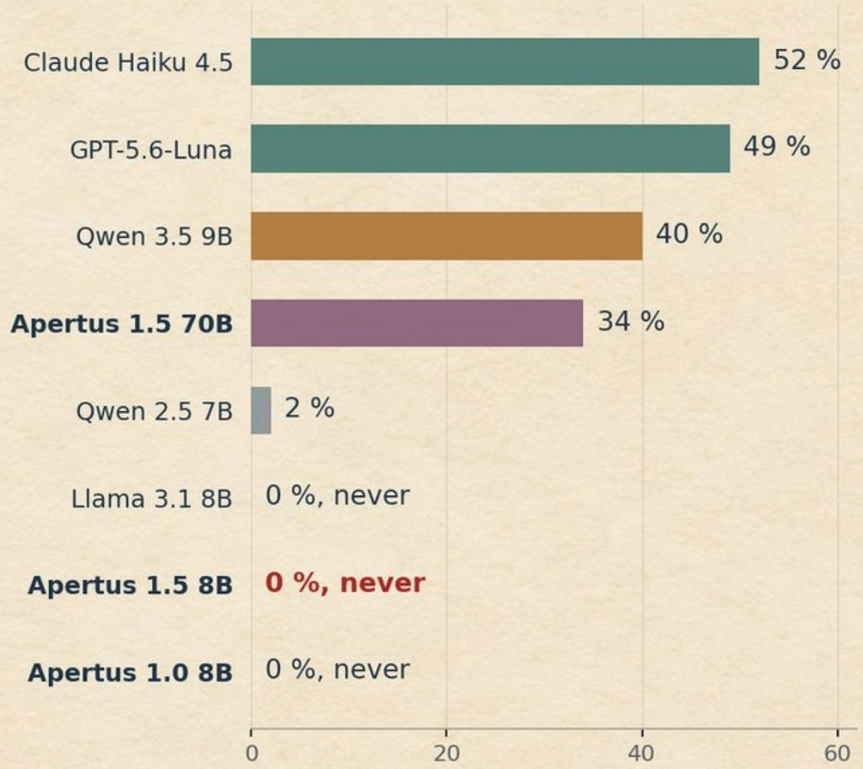
Date absent from test input

Expected: null

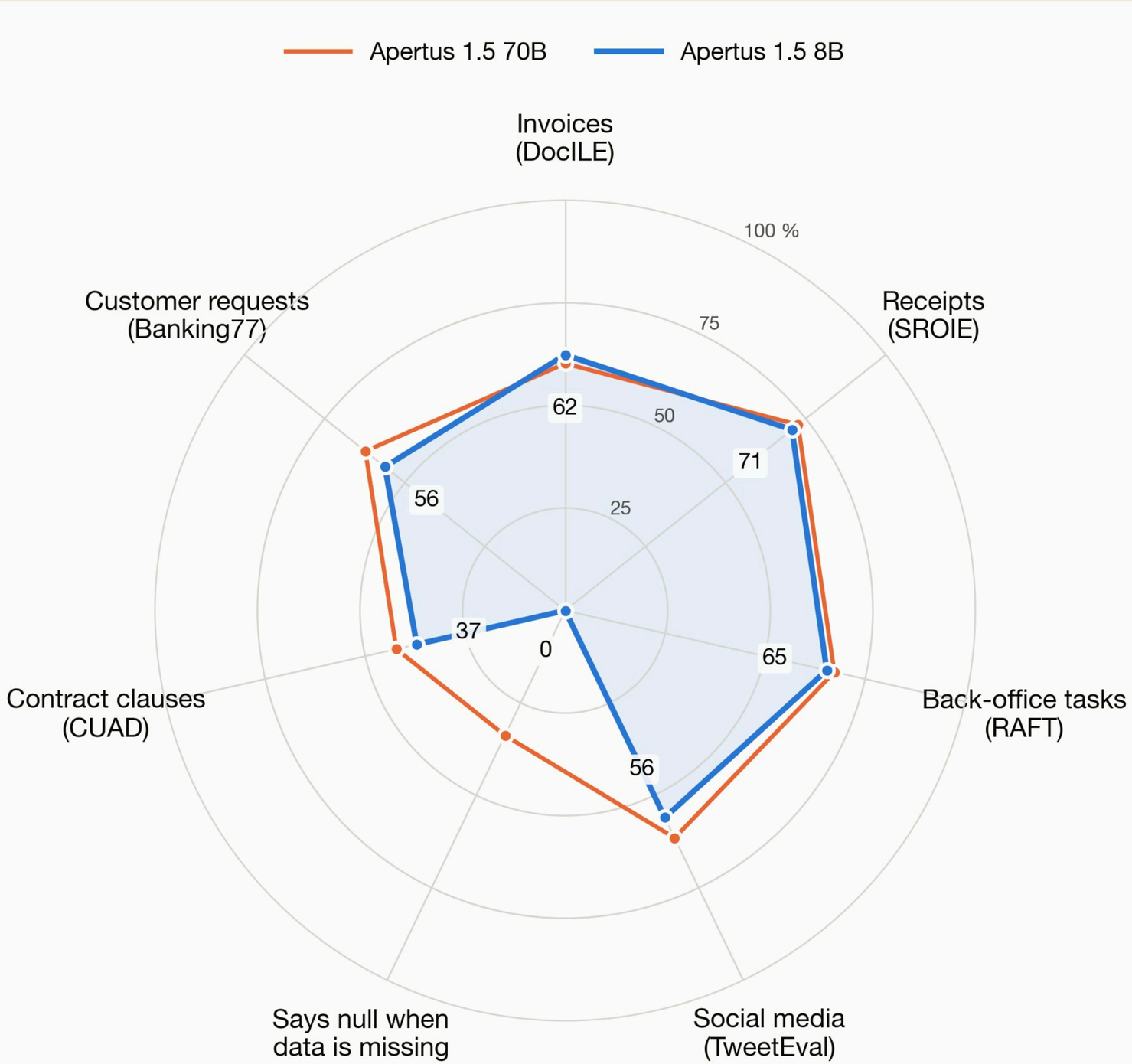
Apertus 1.5 8B	“10/04/2024”	✗ Not in the source
Apertus 1.5 70B	“null”	✓

When the field is missing

80 receipts · date or total removed



For your hack: check every filled field against the source before it is booked.



What the 70B adds over the 8B

70B adds

JUDGEMENT

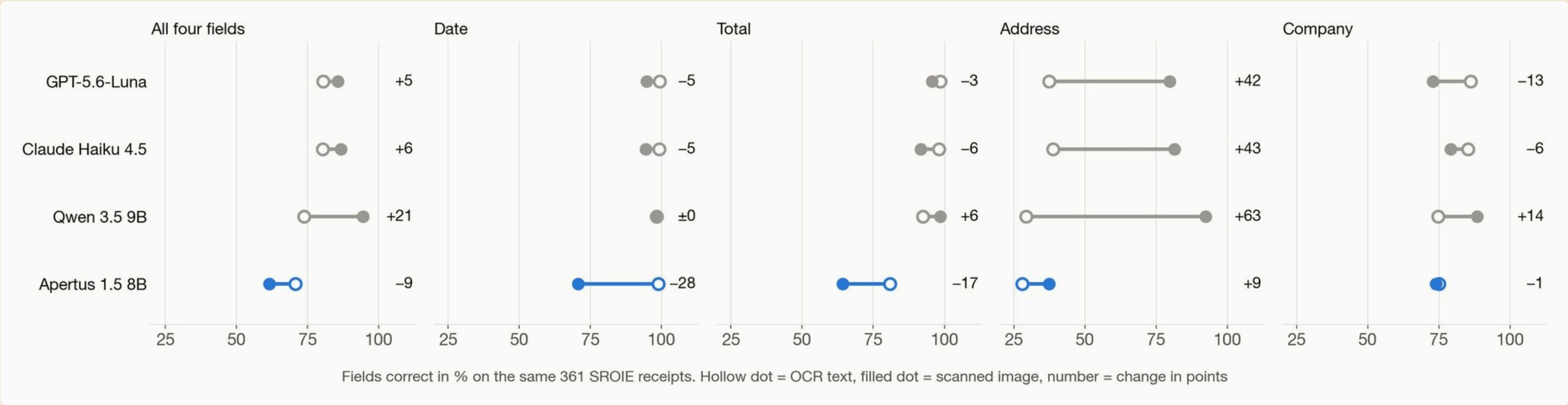
Says «null» (34 % against 0 %), irony 78 against 61, finer contract types.

Same level

EVERYDAY WORK

Invoices, receipts and back-office tasks: about two points apart.

Vision test: the scan instead of the text [8]



APERTUS 1.5 8B · SCAN IS WORSE

71 → 62 % of fields. Misreads digits, 193.00 becomes 198. Never says null.

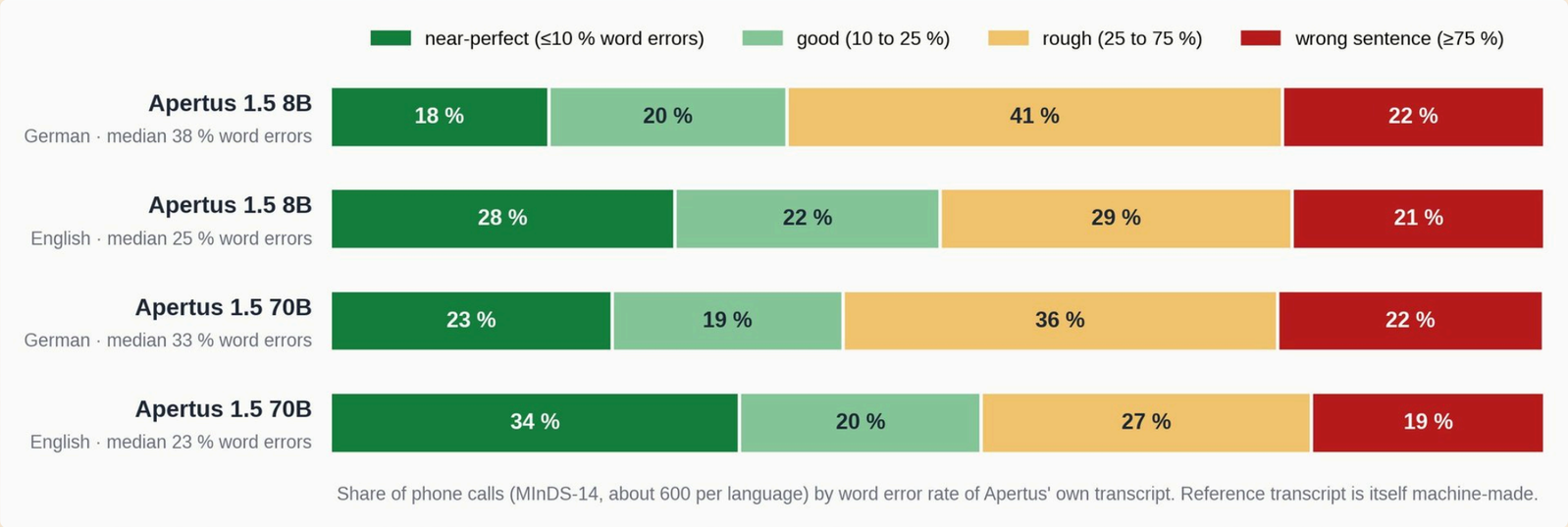
QWEN 3.5 9B · SCAN IS BETTER

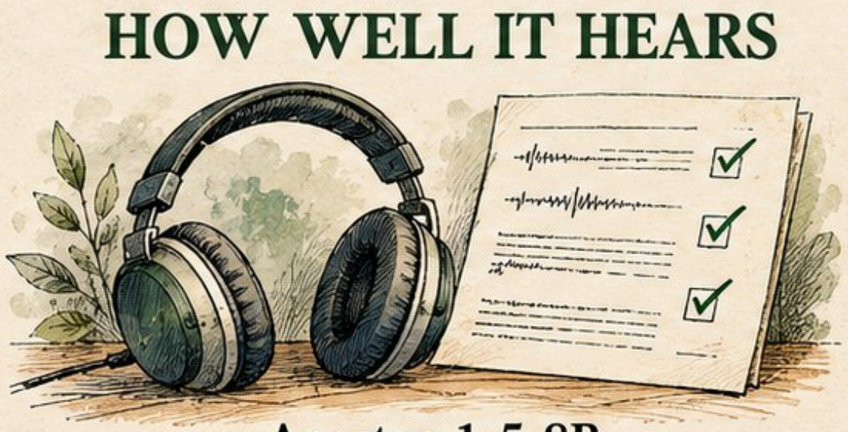
74 → 94 %, address 29 → 93 %. A quarter of the tokens.

FOR YOUR BUILD · TWO STEPS

Read the text first (OCR), then extract from the text. Check every date and total.

Audio test: how well does it hear? [8]





Apertus 1.5 8B
Median word errors
German 38% · English 25%



A fluent sentence nobody said.
Sometimes English for German audio.



Transcribe, check, then decide.
Direct audio-to-intent: 10%

Key Takeaway from Part 2

Apertus 1.5 is a solid document model, 8B and 70B alike. Both read dates and amounts on invoices and receipts reliably, and both run privately in your own infrastructure.

THE 70B ADDS JUDGMENT

Says «null» when data is missing: **34 %** against 0 %. Reads irony better.

Reading itself: about the same level.

Start with the 8B; use the 70B where data can be missing.

SAME WEAK SPOTS · BOTH SIZES

- **Labels:** only from the list
- **Long lists:** two steps
- **Every field:** check the source

IMAGES AND AUDIO · TWO STEPS

First read or transcribe, then decide on the text. In one step: **11 %** of calls right, in two: **74 to 78 %**.

Check every date and total from a scan (tested on the 8B).

The pattern: Apertus does better when you split the work. One thing at a time, first recognise, then decide.

Part 3 · Why should you care?
& How to avoid Pitfalls?

Where our findings

relate to the challenges

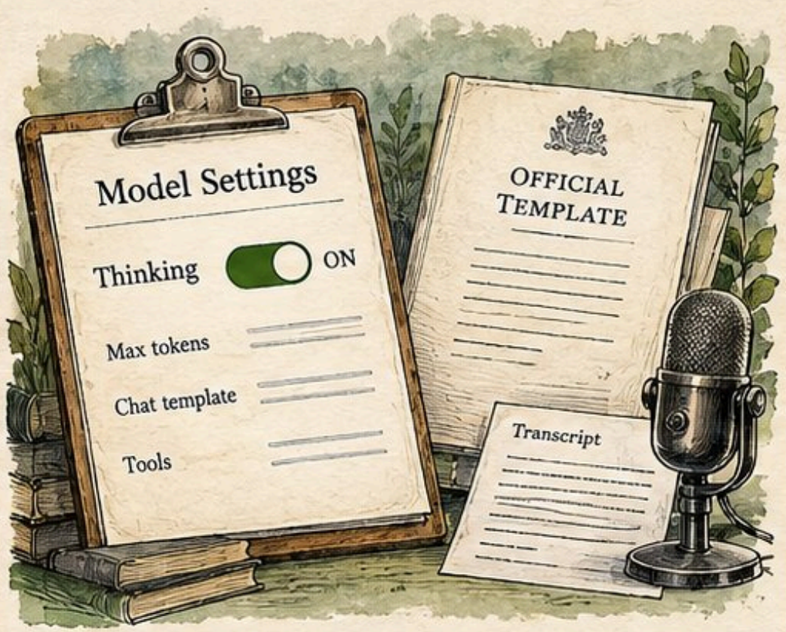


Coloured dot: this finding is a good starting point for that challenge.



DOCUMENTS AND LABELS

- ☑ **Look-alike labels**
  Merge, or give one example each
- ☑ **Long lists**
  Two steps; only allowed labels
- ☑ **Roles**
  Name the document type
- ☑ **Missing data**
  Allow null; check against the source
- ☑ **Scans**
  Verify every date and total



MODEL SETTINGS

- ☑ **Thinking on**
  For multi-step maths
- ☑ **Thinking off**
  For strict formats and tool calls
- ☑ **Token budget**
  2,048 or more with thinking
- ☑ **Chat template**
  Use the official one
- ☑ **Images and audio**
  Read or transcribe first, then decide



BEFORE YOU TRUST A SCORE

- ☑ **Raw outputs**
  Read them before blaming the model
- ☑ **Scoring**
  Compare numbers as numbers, dates as dates
- ☑ **Languages**
  Test your own; dialects are untested
- ☑ **Reproducibility**
  Record revision, template and sampling

Blue: business tasks. **Green:** academic benchmarks and getting-started guide.
Orange: lessons from our evaluation.



ABOUT APERTUS 1.5

Text
Solid middle field, near GPT-3.5 on these tasks.
On invoices, the 8B scores higher.

Specific weak spots
Long label lists, off-list answers.
8B never returned null in our missing-field test.

70B
Adds judgement more than reading.

Images and audio
Weaker in our tests. Read or transcribe first,
then decide on the text.

Thinking
Under 5 points gained on tested pattern tasks,
at 20–50 seconds per case.

Openness
Training data can be inspected.
Possible performance trade-offs remain an open question.



ABOUT MEASURING

Check the reference
OCR text and call transcripts were machine-made
and can contain errors.

Split tasks into steps
Find where the model fails.

Scoring and hosting
Can shift scores more than some model differences.

Disclose differences
Size, quantisation, temperature, modalities
and possible training overlap.

Keep preliminary results preliminary
Vision and audio: one dataset each.

Questions & Contact

University of St.Gallen
Institute for Computer Science

NLP LAB
Chair for Data Science and Natural Language Processing





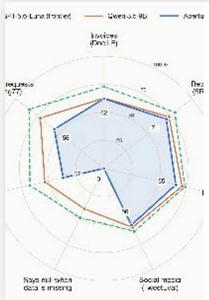
Siegfried Handschuh
siegfried.handschuh@unisg.ch

Discord · sha_hsg

Götz-Henrik Wiegand
goetz-henrik.wiegand@unisg.ch

Discord · gh_wiegand

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Apertus 1.5 Under Test - Real Business Tasks

siegfried handschuh · October 01, 2026

large-language-models · evaluation · benchmarks · vision · multimodal · document-understanding · artificial-intelligence · switzerland

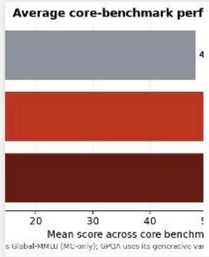
From Exams to the Back Office

Getting Started with Apertus 1.5: Architecture, API, and Local Deployment

götz-henrik wiegand · October 01, 2026

transformers · machine-learning · large-language-models · artificial-intelligence · switzerland

You have an idea for Hack Apertus. Now you need a model endpoint, a working prompt, and a way to connect the two. This post introduces Apertus 1.5 and takes you through the practical steps: calling a hosted API, running the model on your own hardware, and enabling its thinking mode.



LLM Benchmark Evaluation - Apertus 1.5-8B

götz-henrik wiegand · July 29, 2026 · October 01, 2026

transformers · machine-learning · large-language-models · artificial-intelligence · switzerland

The Update

References

[1] Wiegand, G.-H. (2026). LLM Benchmark Evaluation: Apertus 1.5-8B. DS-NLP Lab Blog, 29 July 2026. https://blog.nlp-lab.ai/2026/07/29/Apertus15Bench.html

[2] Casanueva, I., Temčinas, T., Gerz, D., Henderson, M. & Vulić, I. (2020). Efficient Intent Detection with Dual Sentence Encoders. Proceedings of the 2nd Workshop on NLP for Conversational AI. https://huggingface.co/datasets/legacy-datasets/banking77

[3] Huang, Z. et al. (2019). ICDAR2019 Competition on Scanned Receipt OCR and Information Extraction. ICDAR 2019.

[4] Hendrycks, D., Burns, C., Chen, A. & Ball, S. (2021). CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. NeurIPS Datasets and Benchmarks.

[5] Alex, N., Lifland, E., Tunstall, L., Thakur, A., Maham, P., Riedel, C. J., Hine, E., Ashurst, C., Sedille, P., Carlier, A., Noetel, M. & Stuhlmüller, A. (2021). RAFT: A Real-World Few-Shot Text Classification Benchmark. NeurIPS Datasets and Benchmarks. https://huggingface.co/datasets/ought/raft

[6] Barbieri, F., Camacho-Collados, J., Espinosa-Anke, L. & Neves, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. Findings of EMNLP 2020.

[7] Šimsa, Š., Šulc, M., Uřičář, M., Patel, Y., Hamdi, A., Kocián, M., Skalický, M., Matas, J., Doucet, A., Coustaty, M. & Karatzas, D. (2023). DocILE Benchmark for Document Information Localization and Extraction. ICDAR 2023. https://github.com/rossumai/docile

[8] Handschuh, S. (2026). Apertus 1.5 Under Test: Real Business Tasks. Six public benchmarks (Banking77, SROIE, CUAD, RAFT, TweetEval, DocILE), about 3,000 cases per model. DS-NLP Lab Blog, 1 October 2026. https://blog.nlp-lab.ai/2026/10/01/Apertus1-5-BusinessBenchmarks.html

Appendix · The six benchma

Banking77: what does the customer want? [2]

Origin

PolyAI, 2020. 13,083 real customer requests to an online bank, each labelled with one of 77 intents.

Tests

Does the model grasp the concrete reason for a message, when many intents are close neighbours?

Our sample

616 requests, 8 for each of the 77 intents. Metric: accuracy.

Easy

Distinct requests: closing the account, a lost phone, editing personal details.

Hard

Close neighbours such as pending transfer against transfer timing, and labels whose name misleads.

«My card hasn't arrived yet.» → `card_arrival`
«Ordered awhile back, what is the ETA in the US?» → `card_delivery_estimate`
«My refund is missing» → `Refund_not_showing_up`

SROIE: what is on this receipt? [3]

Origin

ICDAR 2019 competition. About 1,000 scanned receipts from shops and restaurants in Malaysia.

Tests

Can the model pull four fields out of noisy OCR text: company, date, address, total?

Our sample

All 361 test receipts, all four fields. Metric: field accuracy and fully correct receipts.

Easy

Date and total: fixed formats, labelled on almost every receipt.

Hard

The address: part of it is often missing from the OCR text. Company names vary in spelling.

`YONG CEN ENTERPRISE 9 40150 SHAH ALAM … DATE: 06/01/2018 … SUB-TOTAL 61.99 … GST 3.72 … TOTAL 65.70`

→ company `YONG CEN ENTERPRISE` · date `06/01/2018` · total `65.70`

CUAD: what does this contract clause do? [4]

Origin

The Atticus Project and UC Berkeley, NeurIPS 2021. 510 real contracts from SEC filings, labelled by lawyers.

Tests

Does the model recognise the legal function of a clause, across 41 clause types?

Our sample

294 clauses: up to 6 per type, plus 60 that match no type. Metric: accuracy.

Easy

Clauses with unmistakable wording: governing law, liquidated damages, most favoured nation.

Hard

Sub-types that look alike: five kinds of licence clause, capped against uncapped liability.

«… governed by the laws and judicial decisions of the State of Florida …» → `Governing Law`
«IN NO EVENT SHALL … TOTAL LIABILITY … EXCEED THE AMOUNTS ACTUALLY RECEIVED …» → `Cap On Liability`
«… a transfer of shareholding … results in any person … gaining Control …» → `Change Of Control`

RAFT: eleven real jobs for an assistant [5]

Origin

Ought and partners, NeurIPS 2021. Eleven classification tasks that organisations actually pay people to do.

Tests

Can the model take on an unfamiliar task from a short description, without training?

Our sample

All 11 tasks, 50 labelled cases each, zero-shot. Metric: macro-F1, as in RAFT.

Easy

Explicit cues: a complaint in a tweet, «overruled» in a court sentence.

Hard

Tasks that need unstated criteria: systematic-review inclusion, reading level, unfair terms.

Customer tweets: «@TopmanAskUs please just give me my money back.» → `complaint`
Court rulings: «accordingly, it is overruled.» → `overruling`
Terms of service: «We may revise these terms from time to time.» → `potentially unfair`

TweetEval: reading between the lines [6]

Origin

Cardiff University and Snap, EMNLP 2020. Seven SemEval tasks on tweets, unified in one format.

Tests

Can the model read short, informal text: irony, emotion, stance, offence, hate, sentiment, emoji?

Our sample

All 7 tasks, 150 test tweets each. Official metric per task, mostly macro-F1.

Easy

Emotion and offensive language, usually carried by strong words.

Hard

Picking one of 20 similar emojis, and stance when the topic is only implied.

Irony: «Leaving whilst its dark is fun. #not» → `irony`
Emotion: «Just ran out of data… #shiver» → `sadness`
Stance on atheism: «Let Your fear come upon every soul in my city» → `against`

DocILE: fill in the invoice form [7]

Origin

Rossum with Czech Technical University and partners, ICDAR 2023. About 6,700 annotated business documents.

Tests

Can the model read who billed whom, which IDs, dates and amounts, from messy OCR text?

Our sample

150 validation documents, 15 fields, text from the supplied OCR. Metric: field accuracy.

Easy

Issue date and tax amount: labelled and in a predictable place.

Hard

Customer addresses over several lines, and telling customer, order and document numbers apart.

`AAA FLAG & BANNER MFG. CO., INC. … INVOICE DATE 10/29/99 … NET 30 DAYS … TOTAL 4,059.38`

→ vendor `AAA FLAG & BANNER MFG. CO., INC.` · issued `10/29/99` · terms `NET 30 DAYS` · total `4,059.38` · due date `null`